CS215 – FINAL PROJECT REPORT

Relationship Between Sleep, Study Habits, Lifestyle Factors and Student Academic Performance

Group 26

1. Project Information

Project Title	Relationship Between Sleep, Study Habits, Lifestyle Factors and Student Academic Performance
Group ID	26
Mentor Name	Namish Bansal

Group Members

lightblue **S. No.**	**Name**	**Roll Number**
1	Chetan	25b0942
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3	Mukesh	25b1009
4	Rishanth	25b1044

Project Links

GitHub Repository	Sinde-Chetan/DAI_CS215_Project
Video Presentation Link	Presentation Video drive link
LLM Usage Links	gemini Link1 , gemini Link2

GitHub Commit

GitHub Initial Commit	first commit

Honor Code and Discussions

We declare that we have not discussed or shared any details, ideas, findings, or information regarding this project with anyone outside our group. All discussions, contributions, and work related to this project have remained strictly within our group.

– chetan , karthik , mukesh , rishanth .

Contribution Breakdown

Team Contributions:

- **Chetan 100%** : **Data Cleaning** and **EDA notebook**, contributed in doing final report pdf file, correcting all regression and other scripts, and in final video presentation explaining regression modeling and validation.
- **Rishanth 100%** : **Analysis** and **Validation**, contributed in doing final report pdf file, correcting all regression and other scripts, and in final video presentation explaining correlation heatmaps and regression plots.
- **Mukesh 100%** : **Statistical Modeling**, contributed in building **Linear** and **Multivariate** regression and plotting graphs of four models M1, M2, M5, M6, and also contributed in final video presentation

explaining data cleaning process and data analysis. Made final team presentation slides.

- **Karthik 100%** : **Statistical Modeling**, contributed in building **Linear** and **Multivariate** regression and plotting graphs of four models M1, M2, M5, M6, and also contributed in final video presentation explaining problem statement and method we followed for project and edited the final video presentation.
- **LLM 100%** : Contributed in making us learn libraries, functions in python, and repeated verification of scripts, and also contributed in writing latex reports.

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1. Technical Content

1.1 Problem Statement

Our project studies how sleep, study hours and daily lifestyle habits are related to students' academic performance.

Our main question is:

How do sleep, study hours and lifestyle habits affect GPA, and which factor has the strongest relationship with it?

Our original proposal also included screen time. Since the dataset did not contain this information, we focused on the available variables.

We tested four main hypotheses:

- **H1:** Study and sleep hours have a positive relationship with GPA.
- **H2:** More social time and less physical activity are associated with lower GPA.
- **H3:** Sleep changes the relationship between study hours and GPA.
- **H4:** A model with multiple lifestyle factors predicts GPA better than study hours alone.

1.2 Dataset and Preprocessing

We used a dataset containing **2,000 student records**. It included study, sleep, extracurricular, social and physical activity hours, GPA and stress level.

GPA ranged from 2.24 to 4.00. Stress levels were Low (297), Moderate (674) and High (1029).

Using Python, we checked for missing values, duplicates and invalid entries. None were found. Stress levels were converted into numbers: Low = 1, Moderate = 2 and High = 3.

1.3 Tools Used

- **Pandas and NumPy:** Data cleaning and calculations.
- **Statsmodels and Scikit-learn:** Regression, hypothesis testing and model comparison.
- **Matplotlib and Seaborn:** Graphs, heatmaps and coefficient plots.

1.4 India Focus

The study is relevant to Indian college students who balance academic work, competitive exams, sleep and other daily activities. Understanding these factors can help students manage their time and studies better.

1.5 Challenges and Exploratory Analysis

1. Daily hours adding up to 24:

The five activity variables always added up to exactly 24 hours. Including all five in regression along with a constant caused perfect multicollinearity. We removed social hours and used it as the reference activity.

2. Missing screen-time information:

Screen time was part of our original proposal but was not available in the dataset. We therefore focused on the other lifestyle variables.

Exploratory results:

The initial correlation analysis showed:

- Study hours had a strong positive correlation with GPA ($r = 0.73$).
- Sleep hours had almost no correlation ($r = 0.00$).
- Physical activity had a negative correlation ($r = -0.34$).
- Stress had a positive correlation ($r = 0.55$).

Stress and physical activity were also related to study hours. Study hours explained around 55% of the

variation in stress. High-stress students studied 8.4 hours daily on average, compared to 5.5 hours for low-stress students.

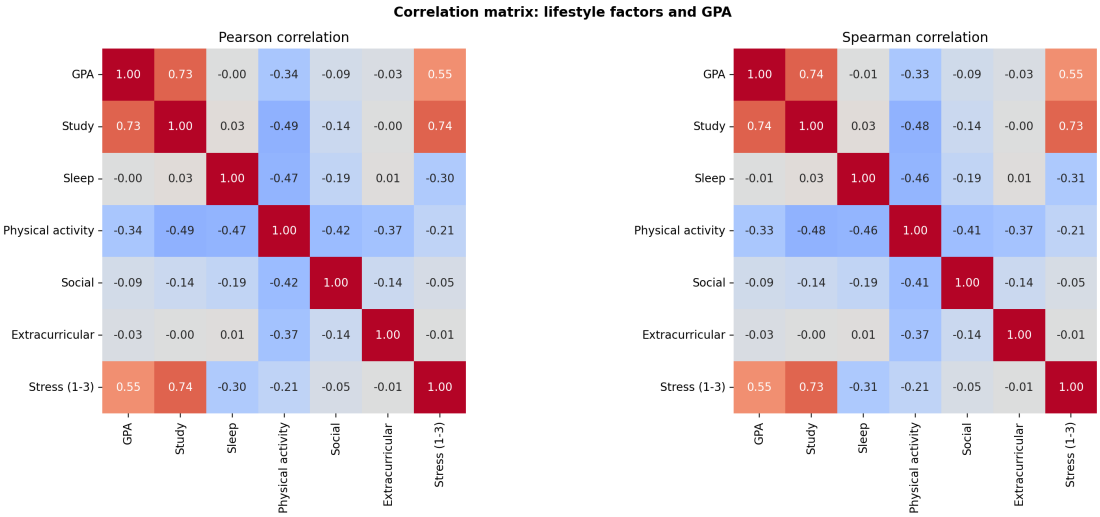


Figure 1: Pearson and Spearman correlation heatmaps.

1.6 Regression Models and Results

We used Ordinary Least Squares (OLS) regression to study the relationship between different factors and GPA. Four models were compared.

Table 1: Comparison of regression models ($n = 2000$).

Model	**Adj.** R^2	**AIC**	**BIC**	**CV** R^2
M1: Study only	0.5392	-705.5	-694.3	0.5370
M4: Study + Sleep	0.5396	-706.0	-689.2	0.5371
M5: Study × Sleep	0.5401	-707.2	-684.8	–
M6: Full model	0.5398	-704.1	-670.5	0.5365

The models had very similar R^2 values. Adding more lifestyle variables did not significantly improve prediction.

Full model (M6):

The model included study, sleep, physical activity, extracurricular hours and stress. Social hours were used as the reference category.

Table 2: Full model regression results.

Variable	**Coef.**	**Std. Error**	p**-value**	**Std. Coef.**
Study	+0.1530	0.0057	$< 10^{-16}$	+0.729
Sleep	-0.0058	0.0041	0.157	-0.029
Physical activity	-0.0013	0.0028	0.638	-0.011
Extracurricular	-0.0088	0.0045	0.051	-0.034
Stress	+0.0002	0.0105	0.984	+0.001

Study hours had the strongest relationship with GPA. An additional hour of study was associated with an increase of approximately 0.153 in GPA, keeping other variables constant. The other variables did not show statistically significant effects at the 5% level.

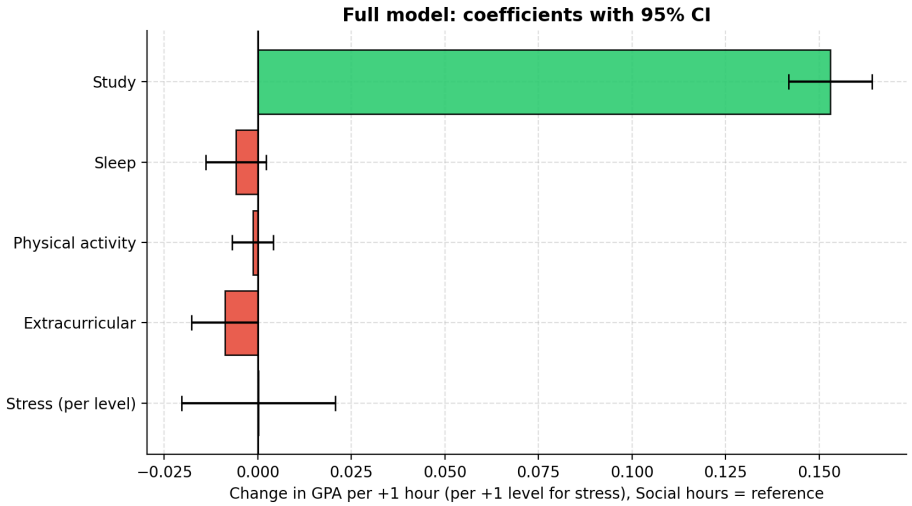


Figure 2: Regression coefficients with 95% confidence intervals.

1.7 Hypothesis Testing

- **H1 – Partially Supported:** Study hours had a significant positive relationship with GPA ($p < 10^{-16}$), but sleep did not ($p = 0.157$).
- **H2 – Not Supported:** The negative relationships of social time and physical activity were not significant after controlling for other variables.
- **H3 – Not Supported:** The study-sleep interaction was not significant at 5% ($p = 0.072$), and its improvement in R^2 was very small.
- **H4 – Not Supported:** Additional lifestyle variables did not significantly improve prediction over study hours alone ($F = 1.66$, $p = 0.158$).

Overall, study hours were the strongest factor associated with GPA in our dataset.

2. Frequently Asked Questions (FAQ)

1. **[Objective] Why did we remove social hours from the regression model?**
   Answer: All five daily activity variables added up to exactly 24 hours. Including all five with a constant caused perfect multicollinearity. We removed social hours and used it as the reference category.
2. **[Objective] Did stress level independently affect GPA?**
   Answer: No significant effect was found. Although stress initially showed a positive correlation with GPA ($r = 0.55$), its regression coefficient became almost zero after considering other variables. This suggests that its initial relationship was largely explained by study hours.
3. **[Subjective] Why did sleep hours not show a significant relationship with GPA?**
   Answer: Sleep duration in our dataset was limited to 5–10 hours. Also, sleep quality and sleeping schedules may matter more than total sleep duration, but our dataset did not include this information.
4. **[Subjective] What is the main takeaway for students?**
   Answer: Study hours showed the strongest relationship with GPA. Students should focus on consistent and effective studying. However, proper sleep, physical activity and a balanced lifestyle are also important for overall well-being.

3. Reflection Essay

Working on this project helped us understand how data analysis works in real situations. Initially, we expected the dataset to contain all the variables mentioned in our proposal. However, screen time was missing, so we had to adjust our analysis.

One of our biggest challenges was understanding why our regression model was not working correctly. We discovered that all five daily activity variables added up to exactly 24 hours, causing perfect multicollinearity. Solving this helped us understand regression models and why some variables need to be removed.

We also learned the difference between correlation and regression. Some variables, such as physical activity and stress, showed relationships with GPA initially. However, these relationships became insignificant when we considered study hours and other factors together.

The project also gave us practical experience in data cleaning, visualization, hypothesis testing and com-

paring regression models.

Overall, we learned that data analysis is not just about generating graphs or running code. It is also about understanding the data, identifying problems and making conclusions based on statistical evidence.